

Viktor Jahnke

CONTACT INFORMATION	ICTP South American Institute for Fundamental Research Instituto de Física Teórica – Universidade Estadual Paulista Rua Dr. Bento Teobaldo Ferraz 271, Bloco II 01140-070 São Paulo – SP, Brazil	viktorjahnke@gmail.com, v.jahnke@unesp.br
RESEARCH INTERESTS	String Theory, Gauge/Gravity Duality, Quantum Chaos, Quantum Field Theory.	
EDUCATION	University of São Paulo , São Paulo, Brazil	
	Ph.D. in Physics, July 2016	
	• Thesis Topic: <i>Applications of the AdS/CFT correspondence to the quark-gluon plasma</i> • Advisor: Diego Trancanelli	
	M.Sc. in Physics, Aug 2010	
	• Topic: <i>Cross sections of K-shell ionization by electron impact, measured from threshold to 100 keV, for Au and Bi.</i> • Advisor: Vito R. Vanin	
	B.Sc. in Physics, July 2010	
	• Final grade: 8.7/10	
CURRENT RESEARCH	My research focuses on the interplay between chaos and the physics of black holes, which, in a sense, can be thought of as maximally chaotic objects. Specifically, I explore this connection within the context of the AdS/CFT correspondence, which posits a duality between a theory of gravity in an Anti-de Sitter (AdS) space and a conformal field theory (CFT) defined on the boundary of that space. This framework offers a powerful tool for investigating quantum gravity and strongly correlated systems.	
RESEARCH EXPERIENCE	Postdoctoral Fellowship IFT–UNESP, São Paulo, Brazil Supervisor: Nathan Berkovits	September 2025 – present
	Postdoctoral Fellowship Gwangju Institute of Science and Technology, Supervisor: Keun-Young Kim	November 2018 – August 2025
	Postdoctoral Fellowship Nuclear Sciences Institute, National Autonomous University of Mexico Supervisor: Alberto Güijosa	November 2016 to August 2018
	Research Assistant (Ph.D) Institute of Physics, University of São Paulo Supervisor: Diego Trancanelli	December 2012 to October 2016
	Research Assistant (master's degree) Institute of Physics, University of São Paulo Supervisors: Vito R. Vanin	August 2010 to August 2012
	Research Assistant (scientific initiation) Institute of Physics, University of São Paulo Supervisors: Vito R. Vanin	December 2006 to July 2010

FUNDING

- **CNPq “Fixação de Talentos” Fellowship (Process No. 446326/2024-0)**
September 2025 – August 2029

Host Institution: ICTP South American Institute for Fundamental Research (ICTP-SAIFR) / Instituto de Física Teórica – UNESP, São Paulo, Brazil

Project Title: “Caos, Informação Quântica e a Correspondência AdS/CFT”

Funded by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Brazil

- **NRF Japan-Korea Human Resource Project** August 2024 – July 2025

Project Title: “Fundamental Theory of Quantum Information: Quantum Entanglement and Quantum Complexity”

Principal Investigators: Keun-Young (Korea’s side), Tadashi Takayanagi (Japan’s side)

- **Korea National Research Foundation Grant** June 2023 – August 2025

Project Title: “Quantum Information and the AdS/CFT Correspondence”

Amount: 150,000,000 South Korean Won

Funded by the Ministry of Education RS-2023-00248186

- **Korea National Research Foundation Grant** July 2020 – June 2023

Project Title: “Quantum Chaos and the AdS/CFT Correspondence”

Amount: 150,000,000 South Korean Won

Funded by the Ministry of Education NRF-2020R1I1A1A01073135

REFEREED JOURNAL PUBLICATIONS

1. Espíndola, Ricardo; **Jahnke, Viktor**; Kim, Keun-Young. “Islands and traversable wormholes,” *Journal of High Energy Physics*, v. 2026, p. 035, 2026.
2. Aguilar-Gutierrez, Sergio E.; Camargo, Hugo A.; **Jahnke, Viktor**; Kim, Keun-Young; Nishida, Mitsuhiro. “Krylov Operator Complexity in Holographic CFTs: Smeared Boundary Reconstruction and the Dual Proper Radial Momentum,” *Physical Review D*, v. 112, p. 126014, 2025.
3. **Jahnke, Viktor**; Nandy, Pratik; Pal, Kuntal; Camargo, Hugo A.; Kim, Keun-Young. “Free Probability Approach to Spectral and Operator Statistics in Rosenzweig–Porter Random Matrix Ensembles,” *Journal of High Energy Physics*, v. 2025, p. 002, 2025.
4. Camargo, Hugo A.; Fu, Yichao; **Jahnke, Viktor**; Pal, Kuntal; Kim, Keun-Young. “Quantum signatures of chaos from free probability,” *Journal of High Energy Physics*, v. 2025, p. 138, 2025.
5. Camargo, Hugo A.; Fu, Yichao; **Jahnke, Viktor**; Kim, Keun-Young; Pal, Kuntal. “Higher-Order Krylov State Complexity in Random Matrix Quenches,” *Journal of High Energy Physics*, v. 07, p. 182, 2025.
6. Camargo, Hugo A.; Huh, Kyoung-Bum; **Jahnke, Viktor**; Jeong, Hyun-Sik; Kim, Keun-Young; Nishida, Mitsuhiro. “Spread and Spectral Complexity in Quantum Spin Chains: From Integrability to Chaos,” preprint (2024).
7. Ahn, Yongjun; **Jahnke, Viktor**; Jeong, Hyun-Sik; Ji, Chang-Woo; Kim, Keun-Young. “On pole-skipping with gauge-invariant variables in holographic axion theories,” *Journal of High Energy Physics*, v. 06, p. 020, 2024.

8. Camargo, Hugo A.; **Jahnke, Viktor**; Jeong, Hyun-Sik; Kim, Keun-Young; Nishida, Mitsuhiro. “Spectral and Krylov Complexity in Billiards Systems,” *Physical Review D*, v. 109, no. 4, p. 046017, 2024. **Selected as Physical Review D Editor’s choice for February of 2024.**
9. Ahn, Byoungjoon; Bak, Sang-Eon; **Jahnke, Viktor**; Kim, Keun-Young. “Traversable wormholes via a double trace deformation involving $U(1)$ conserved current operators,” *Physical Review D*, v. 109, no. 6, p. 066016, 2024.
10. Camargo, Hugo A.; **Jahnke, Viktor**; Kim, Keun-Young; Nishida, Mitsuhiro. “Krylov Complexity in Free and Interacting Scalar Field Theories with Bounded Power Spectrum,” *Journal of High Energy Physics*, v. 05, p. 226, 2023.
11. Pedraza, Juan; de Boer, Jan; **Jahnke, Viktor**; Kim, Keun-Young. “Worksheet traversable wormholes,” *Journal of High Energy Physics*, v. 05, p. 141, 2023.
12. Ahn, Byoungjoon; Ahn, Yongjun; Bak, Sang-Eon; **Jahnke, Viktor**; Kim, Keun-Young. “Holographic teleportation in higher dimensions,” *Journal of High Energy Physics*, v. 07, p. 219, 2021.
13. Huh, Kyoung-Bum; Ikeda, Kazuki; **Jahnke, Viktor**; Kim, Keun-Young. “Diagnosing first- and second-order phase transitions with probes of quantum chaos,” *Physical Review E*, v. 104, no. 2, p. 024136, 2021.
14. Ahn, Yongjun; **Jahnke, Viktor**; Jeong, Hyun-Sik; Kim, Keun-Young; Lee, Kyung-Sun; Nishida, Mitsuhiro. “Classifying pole-skipping points,” *Journal of High Energy Physics*, v. 03, p. 175, 2021.
15. Ahn, Yongjun; **Jahnke, Viktor**; Jeong, Hyun-Sik; Kim, Keun-Young; Lee, Kyung-Sun; Nishida, Mitsuhiro. “Pole-skipping of scalar and vector fields in hyperbolic space: conformal blocks and holography,” *Journal of High Energy Physics*, v. 09, p. 111, 2020.
16. Ahn, Yongjun; **Jahnke, Viktor**; Jeong, Hyun-Sik; Kim, Keun-Young. “Scrambling in Hyperbolic Black Holes: shock waves and pole-skipping,” *Journal of High Energy Physics*, v. 10, p. 257, 2019.
17. **Jahnke, Viktor**; Kim, Keun-Young; Yoon, Junggi. “On the Chaos Bound in Rotating Black Holes,” *Journal of High Energy Physics*, v. 05, p. 037, 2019.
18. **Jahnke, Viktor**. “Recent developments in the holographic description of quantum chaos,” *Advances in High Energy Physics*, v. 2019, p. 9632708, 2019.
19. Mansoori, Seyed Ali H.; **Jahnke, Viktor**; Qaemmaqami, Mohammad M.; Olivas, Yaithd D. “Holographic complexity of anisotropic branes,” *Physical Review D*, v. 100, no. 4, p. 046014, 2019.
20. Fischler, Willy; Pedraza, Juan F.; **Jahnke, Viktor**. “Chaos and entanglement spreading in a non-commutative gauge theory,” *Journal of High Energy Physics*, v. 11, p. 072, 2018; erratum in JHEP 02 (2021) 149.
21. Ávila, Daniel; **Jahnke, Viktor**; Patiño, Leonardo. “Chaos, Diffusivity, and Spreading of Entanglement in Magnetic Branes, and the Strengthening of the Internal Interaction,” *Journal of High Energy Physics*, v. 1809, p. 131, 2018.
22. **Jahnke, Viktor**. “Delocalizing Entanglement of Anisotropic Black Branes,” *Journal of High Energy Physics*, v. 1801, p. 102, 2018.

23. **Jahnke, Viktor**; Misobuchi, Anderson Seigo. “Probing strongly coupled anisotropic plasmas from higher curvature gravity,” *European Physics Journal C*, v. 76, no. 6, p. 309, 2016.
24. **Jahnke, Viktor**; Misobuchi, Anderson Seigo; Trancanelli, Diego. “Holographic renormalization and anisotropic black branes in higher curvature gravity,” *Journal of High Energy Physics*, v. 2015, p. 122, 2015.
25. **Jahnke, Viktor**; Luna, Andrés; Patiño, Leonardo; Trancanelli, Diego. “More on thermal probes of a strongly coupled anisotropic plasma,” *Journal of High Energy Physics*, v. 2014, p. 10.1007/JHEP01, 2014.
26. **Jahnke, Viktor**; Misobuchi, Anderson Seigo; Trancanelli, Diego. “Chern-Simons diffusion rate from higher curvature gravity,” *Physical Review D: Particles, Fields, Gravitation, and Cosmology*, v. 89, p. 107901, 2014.
27. Fernandez-Varea, Jose M.; **Jahnke, Viktor**; Maidana, Nora L.; Malafronte, Alexandre A.; Vanin, Vito R. “Cross sections of K-shell ionization by electron impact, measured from threshold to 100 keV, for Au and Bi,” *Journal of Physics B: Atomic, Molecular and Optical Physics*, v. 47, p. 155201, 2014.
28. Maidana, Nora L.; Vanin, Vito R.; **Jahnke, Viktor**; Fernandez-Varea, Jose M.; Martins, Marcos N.; Brualla, Lorenzo. “Efficiency calibration of x-ray HPGe detectors for photons with energies above the Ge K binding energy,” *Nuclear Instruments and Methods in Physics Research Section A*, v. 729, p. 371-380, 2013.

AWARDS

- Gwangju Institute of Science and Technology Excellence in Research Award 2022
- Student Awards — Institute of Theoretical Physics of UNESP: Honorable mention for the 8th place in the “V Prêmio FIFT para Jovens Físicos” 2009

TEACHING EXPERIENCE

Teaching Assistant	August - December of 2015
4300458 - Classical Mechanics	
Instructor: Vito R. Vanin	
Department of Experimental Physics,	
University of São Paulo	
Teaching Assistant	March - July of 2015
4300307 - Mathematical Physics II	
Instructor: João C. A. Barata	
Department of Mathematical Physics,	
University of São Paulo	
Teaching Assistant	August - December of 2014
PGF5001 - Quantum Mechanics I	
Instructor: Diego Trancanelli	
Department of Mathematical Physics,	
University of São Paulo	
Teaching Assistant	March - July of 2014
4300211 - Physics III (Eletromagnetism)	
Instructor: José L. M. Duarte	
Department of Nuclear Physics,	
University of São Paulo	

- **Seminar:** “Quantum Signatures of Chaos from Free Probability”
Date: September 12, 2024
Location: Okinawa Institute of Science and Technology (OIST), QUASt Seminar Series, online
Presentation Title: “Quantum Signatures of Chaos from Free Probability”
Recording: <https://www.youtube.com/watch?v=h4coT5asmNI&t=10s>
- **Seminar:** “Quantum Signatures of Chaos from Free Probability”
Date: April 14, 2025
Location: Kavli Institute for Theoretical Sciences (KITS), University of Chinese Academy of Sciences (UCAS), Beijing, China (online)
Presentation Title: “Quantum Signatures of Chaos from Free Probability”
- **Conference:** “Quantum Gravity and Information in Expanding Universe”
Date: February 17, 2025
Location: Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Kyoto, Japan
Presentation Title: “Operator Spectral Statistics, Chaos, and Asymptotic Freeness in Quantum Many-Body Systems”
Link: indico.yukawa.kyoto-u.ac.jp/event/52
- **Conference:** “Quantum Gravity of Open Systems”
Date: February 6, 2025
Location: Instituto de Física Teórica (IFT-UAM/CSIC), Madrid, Spain
Presentation Title: “Operator Spectral Statistics, Chaos, and Asymptotic Freeness in Quantum Many-Body Systems”
- **Conference:** “Gauge/Gravity Duality 2024”
Date: November 30 – December 4, 2024
Location: Sanya, Hainan, China
Presentation Title: “Operator Spectral Statistics, Chaos, and Asymptotic Freeness in Quantum Many-Body Systems”
Link: indico.itp.ac.cn/event/257
- **Conference:** “Frontiers of Holographic Duality - 5”
Date: December 8
Location: Moscow, Steklov Mathematical Institute, online
Presentation Title: “Krylov Complexity in Free and Interacting Scalar Field Theories with Bounded Power Spectrum”
- **Conference:** “Holography 2023: quantum matter and spacetime”
Date: January 16-24, 2023 (in-person)
Location: Asian Pacific Center for Theoretical Physics (APCTP), Pohang
Presentation Title: “Krylov complexity in QFTs”
- **Conference:** “Integrability, Deformations and Chaos”
Date: July 25-27, 2023 (in-person)
Location: Okinawa Institute of Science and Technology
Presentation Title: “Krylov complexity in interacting QFTs”
- **Conference:** “Frontiers of Holographic Duality-4”
Date: December 2022
Location: Steklov Mathematical Institute, Moscow (online)
Presentation Title: “Traversable wormholes and conservation laws”
- **Conference:** “East Asia Joint Workshop on Fields and Strings 2022”
Date: November 2022
Location: Korean Institute for Advanced Study, Seoul (hybrid)
Presentation Title: “Comments about holographic teleportation”
- **Conference:** “Holography 2022: quantum matter and spacetime”
Date: August 2022
Location: Asian Pacific Center for Theoretical Physics (APCTP), Pohang (online)

- Presentation Title:** “Holographic teleportation with conservation laws: diffusion on traversable wormholes”
- **Conference:** “III Siembra Holograv: Young Frontiers Meeting”
Date: August 2022
Location: Amsterdam, Saclay, Bariloche, Valparaíso (online)
Presentation Title: “Holographic teleportation with conservation laws: diffusion on traversable wormholes”
 - **Conference:** “Advances in Theoretical Physics”, Kyung-Hee University - APCTP
Date: January 2022
Location: The Grand Sumorum, Jeju (hybrid)
Presentation Title: “Holographic teleportation with conservation laws: diffusion on traversable wormholes”
 - **Conference:** “Recent developments in the holographic principle”, GIST-APCTP
Date: October 2021
Location: Mokpo
Presentation Title: “Comments about holographic teleportation”
 - **Conference:** “String theory, gravity and cosmology”
Date: November 2020
Location: Asian Pacific Center for Theoretical Physics, Pohang
Presentation Title: “Holographic teleportation protocol in higher dimensions”
 - **Conference:** “APCTP - Quantum Matter and Quantum Information with Holography 2020”
Date: August 2020
Location: Asian Pacific Center for Theoretical Physics, Pohang (online)
Presentation Title: “Scrambling and Pole-skipping in Hyperbolic Black Holes”
 - **Conference:** “2019 Korean Physical Society Fall meeting”
Date: October 2019
Location: Kimdaejung Convention Center, Gwangju
Presentation Title: “The Gravity Dual of OTOCs”

REFERENCES

Prof. Diego Trancanelli

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 Università di Modena e Reggio Emilia
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